

QUANTITATIVE RESEARCH NOTE

BTC–Nasdaq Cross-Market Signal Validation

Leakage-aware event research with realistic MNQ execution

Validation rebuild

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Original coursework and research

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Abstract

This study asks whether BTC-derived variables improve prediction of Nasdaq futures event returns over a comparable NQ-only momentum baseline after realistic costs. The earlier notebook work is rebuilt as a causal event pipeline: features are known at minute t close, execution occurs at the next open, positions do not overlap, discontinuities are rejected, and model choices are made from purged expanding walk-forward predictions. The locked final test contains 824 candidate events. The BTC–NQ rule and Random Forest have positive point estimates, but every 95% block-bootstrap interval crosses zero. The useful result is a reproducible model-validation study, not a claim of deployable alpha.

Executive finding. The prespecified positive-edge criterion is not met. BTC variables produce an interesting positive rule-based estimate, but uncertainty remains large and the cross-market Logistic Regression does not materially improve upon its NQ-only ablation.

1 Research question and decision rule

The candidate universe consists of large 15-minute NQ momentum impulses during the US cash session. Four levels are compared: an NQ momentum baseline, a BTC–NQ agreement rule, Logistic Regression, and a constrained Random Forest. The separate NQ-only Logistic Regression is the principal ablation for incremental BTC information.

A positive edge may be declared only when the locked final test has at least 50 accepted trades, positive net P&L, and a positive lower bound for a 95% day-block bootstrap confidence interval of mean net P&L. Profit factor is capped for reporting and cannot override the minimum-sample rule.

Final strategy	Trades	Gross USD	Net USD	Mean USD	95% CI, mean USD
NQ momentum baseline	824	2,149.50	-322.50	-0.39	[-8.62, 8.49]
BTC–NQ rule	413	4,136.50	2,897.50	7.02	[-5.35, 19.49]
LogReg, NQ only	803	1,782.50	-626.50	-0.78	[-9.21, 8.05]
LogReg, cross-market	742	2,610.50	384.50	0.52	[-8.54, 10.13]
Random Forest, cross-market	55	1,795.00	1,630.00	29.64	[-20.83, 79.09]

Table 1: Locked final-test results for one MNQ contract after the configured round-trip cost.

2 Audit of the legacy backtest

The original notebooks were useful exploratory work, but their headline metrics cannot be interpreted as executable performance. They remain under **legacy/** for attribution and auditability and are not a source of truth.

Material failure modes

- **Costs dominated the apparent edge.** One notebook reported 14,554 NQ points from 258,207 trades, only 0.056 point per trade. A later ML notebook reported 0.075 point per trade. Both averages are below one 0.25-point MNQ tick before commission.
- **Overlapping returns were summed.** Neighboring signal bars could represent the same market move, inflating sample size and making aggregate P&L invalid.
- **Execution used unavailable information.** A close required to form the signal was also used as the fill. The maintained engine enters at the next bar's open.
- **Time and contract handling were unsafe.** Fixed UTC offsets ignored daylight-saving transitions, while rolls and missing bars were not consistently excluded.
- **Selection and evaluation were mixed.** Thresholds and winners could be chosen with final-period information. The rebuild uses pre-2026 OOF predictions for selection and locks the final configuration.
- **Sharpe annualization and ranking were unstable.** Per-trade observations were annualized incorrectly, and an infinite profit factor could elevate configurations with one to three validation trades.

Legacy evidence. Historical searches included alternative windows, horizons, filters, and exits. They explain where several research choices originated, but they do not provide clean confirmatory evidence because the search process was broad and the old backtest mechanics were flawed.

3 Data and economic assumptions

BTCUSDT USD-M one-minute bars are public Binance data. NQ one-minute bars are a local IBKR-derived stitched contract series and are not redistributed. The repository supplies a schema, Binance downloader, and synthetic test fixtures instead of proprietary data.

MNQ economics use the official multiplier of \$2 times the index, a 0.25-point minimum tick, and a \$0.50 tick value; see the [CME product page](#). The locked conservative scenario assumes one tick of slippage and \$1 commission per side, giving a \$3.00 round trip. Fees remain configurable and should be checked against the current [IBKR futures schedule](#).

4 Causal event protocol

At minute t , every feature uses bars available no later than that bar's close. An event is the first bar on which the absolute 15-minute NQ return crosses the training-fitted 70th-percentile threshold (8.0462 bps). Direction is the sign of that return. Entry is the open of $t + 1$; exit is the open 30 minutes after entry. A global cooldown permits one open position at a time.

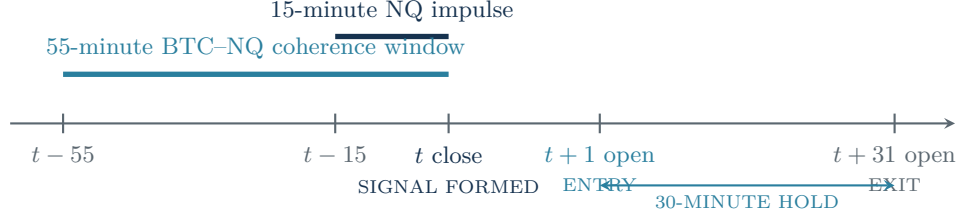


Figure 1: Causal timing. The signal close is never used as the execution price.

Events are rejected when a required feature is missing, a one-minute timestamp is absent, the NQ contract identifier changes, entry and exit do not share a session date, or exit would occur after 16:00 New York time.

Feature sets and labels

The NQ-only set contains 5/15/30-minute momentum, 15/60-minute volatility, 14-minute ATR, RSI, and cyclical time features. The cross-market set adds BTC momentum and volatility, exact rolling 55-minute Spearman correlation, a 55-minute directional hit ratio, and the BTC-minus-NQ momentum spread. For direction $d_i \in \{-1, 1\}$,

$$\text{points}_i = d_i(P_i^{\text{exit}} - P_i^{\text{entry}}).$$

Gross USD equals points times \$2; net USD subtracts the round-trip cost. The label is one only when net USD is positive.

Interpretation. The maintained experiment tests whether BTC features improve selection of a predefined NQ momentum event. It does not establish a structural BTC-to-NQ causal channel, and it does not directly test a dynamic exit after correlation decay.

5 Research specification and parameter provenance

Not every number has the same epistemic status. Market mechanics are imposed by execution; train-fitted quantities are estimated without final-test observations; frozen heuristics encode the historical hypothesis and remain potential sources of specification risk.

Parameter	Status	Rationale and limitation
15-minute NQ momentum	FROZEN HEURISTIC	Short-horizon impulse definition inherited from legacy 15/30-minute experiments. It is interpretable, but the maintained pipeline does not prove 15 minutes is optimal.
70th percentile impulse	TRAIN-FITTED	Defines an above-typical move while retaining an analyzable event count. The numerical threshold is fitted before validation; the percentile itself is a frozen design choice.
55-minute coherence	FROZEN HEURISTIC	Inherited from prior research that compared 55- and 90-minute windows. It balances recency and sample size but has no unique theoretical claim.
Spearman ≥ 0.30	FROZEN HEURISTIC	A pragmatic moderate-positive-association cutoff, fixed before final evaluation. It is not an estimated economic boundary.
Hit ratio ≥ 0.60	FROZEN HEURISTIC	Requires same-sign one-minute moves in a clear majority of the window. It is not statistically identified as an optimal cutoff.
30-minute fixed hold	FROZEN HEURISTIC	Represents short-lived continuation and came from earlier horizon exploration. It is a benchmark, not a validated optimal exit.
Minimum 50 trades	RESEARCH GUARDRAIL	Prevents claims based on extremely small accepted samples. It is not a substitute for a formal power calculation.
60-minute purge	VALIDATION CONTROL	Exceeds the 30-minute label horizon and separates training outcomes from the next fold boundary.
Next-open fills, session and rolls	MARKET MECHANICS	Enforce causal execution and exclude paths that are not comparable because of closure, discontinuity, or contract change.

Table 2: Parameter provenance. “Frozen” means fixed before the locked final test, not theoretically optimal.

No hidden sensitivity claim. The clean pipeline did not rerun a comprehensive 30/55/90-minute window study or optimize the 0.30/0.60 rule. Legacy comparisons are provenance, not validation. A new sensitivity study must be versioned as a new hypothesis and evaluated on new untouched data.

6 Walk-forward validation and model controls

Expanding quarterly folds begin in 2024Q4. Training observations precede validation by the configured purge. Scaling is fitted inside each Logistic Regression fold. Random Forest permutation importance and Logistic Regression coefficients come from fold-local estimators; final-test observations do not participate.

Probability thresholds in $\{0.45, 0.50, 0.55, 0.60, 0.65, 0.70\}$ are ranked using OOF events only and must retain at least 50 trades. The ledger records every feasible comparison and the manifest reports 20 trials. Selection-bias context follows the [Deflated Sharpe Ratio](#) and [Probability of Backtest Overfitting](#) literature. Reported DSR values are approximate diagnostics, not p-values.

7 Validation evidence

The OOF NQ baseline lost \$6,369.50 over 2,500 events; the BTC-NQ rule lost \$4,183 over 1,187 events. Both Logistic Regression selections were negative after costs. The Random Forest earned \$1,102 over 356 selected validation trades, but its mean-net 95% interval was [-\$7.88, \$14.00]. No selected model established a stable validation edge.

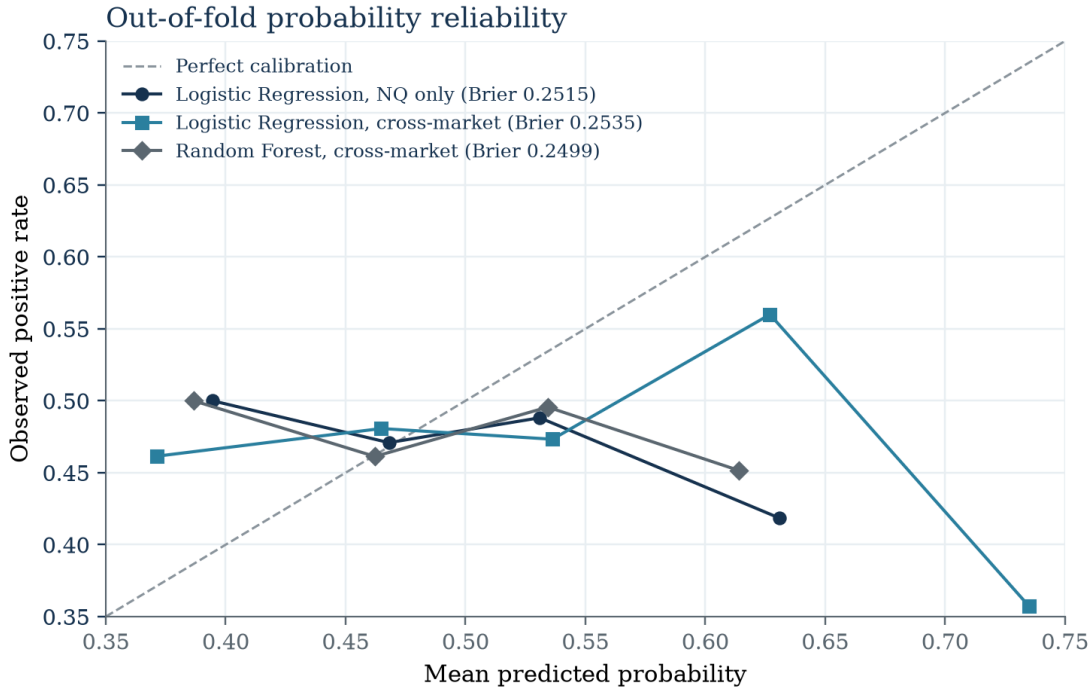


Figure 2: Out-of-fold reliability using bins with at least ten observations. Predictions cluster near 0.5 and do not demonstrate strong discrimination.

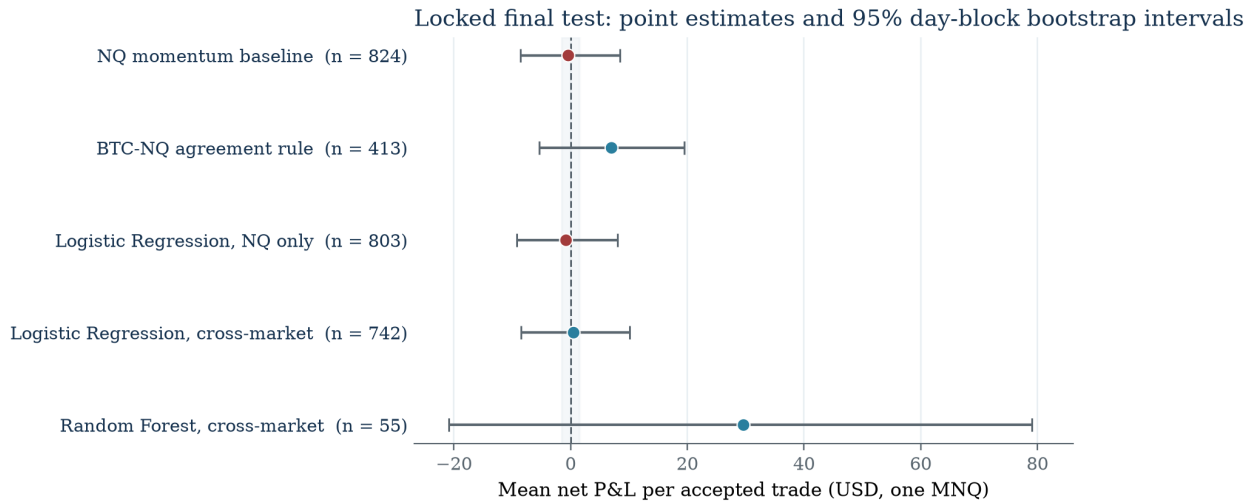
The NQ-only and cross-market Logistic Regression Brier scores are 0.2515 and 0.2535 respectively; the constrained Random Forest score is 0.2499. Cross-market Logistic Regression therefore does not show incremental calibration value over the NQ-only ablation. Coefficients and permutation importance remain predictive diagnostics, not causal estimates.

Locked research state

The manifest stores the sanitized configuration, data range, event threshold, feature lists, OOF selections, trial count, maintained-package SHA-256, and locked-setting hash. The final command rejects a changed configuration or source tree and refuses a second final evaluation without an explicit auditing override.

8 Locked final test

The final interval produced 824 eligible events. Figure 3 reports point estimates together with their uncertainty; this presentation deliberately avoids treating positive bars as evidence of positive population means.



Intervals crossing zero indicate that the sign of the population mean is unresolved.

Figure 3: Mean net P&L and 95% day-block bootstrap intervals. Every interval includes zero.

9 Does the hypothesis have potential?

There is a narrow reason to continue studying the question, but not to trade the result. The BTC-NQ agreement rule earns a positive final-test point estimate of \$7.02 per trade over 413 trades. Its 95% interval, [-\$5.35, \$19.49], still admits both a meaningful loss and a useful gain. That is an unresolved estimate, not confirmed alpha.

The cross-market Logistic Regression does not materially outperform its NQ-only ablation, which weakens a general claim that the chosen BTC feature set adds predictive information. The Random Forest estimate is positive but rests on only 55 accepted trades and has a very wide interval. Together, these results support a follow-up hypothesis about conditional dependence, not deployment.

Balanced conclusion. The point estimate of the interpretable BTC-NQ rule is economically interesting enough to motivate new research. The validation and uncertainty evidence is not strong enough to claim robust after-cost predictability.

10 Limitations

- One stitched NQ source cannot independently verify vendor corrections, roll construction, or historical bid/ask conditions.
- Fixed slippage simplifies spread variation, latency, queue position, partial fills, market impact, and exchange fees.
- The event and feature families reflect prior human exploration. Trial logging and approximate DSR expose but cannot reconstruct all selection decisions.
- The final interval spans roughly four months. Day-block bootstrap preserves within-day dependence but cannot generate unobserved regimes.
- The 30-minute exit and 55-minute coherence window are design choices. The present result cannot distinguish a weak hypothesis from a poorly chosen horizon.
- Feature importance is fold-specific and predictive, not evidence of a structural or causal BTC-to-NQ transmission mechanism.

11 Pre-specified future research

A follow-up should be treated as a new study rather than an extension of the locked test:

1. Compare 30/55/90-minute coherence windows and report the entire sensitivity surface rather than its winner.
2. Test a dynamic exit that closes after the entry coherence condition decays, with next-open execution and a fixed maximum hold.
3. Estimate whether standardized BTC or NQ impulse magnitude has a monotonic relationship with subsequent NQ returns.
4. Report long and short events separately to distinguish cross-market information from directional equity beta.
5. Freeze the new specification before collecting or opening a new untouched evaluation period.

12 Reproducibility and conclusion

```
python -m cross_market.run --config configs/research.yaml --stage validation
python -m cross_market.run --config configs/research.yaml --stage final
```

The project demonstrates causal timing, event construction, futures cost accounting, purged walk-forward selection, calibration, drift diagnostics, and an immutable final-test audit trail. **The hypothesis that the current BTC features create a robust after-cost NQ edge is not established by this sample.**